ch1 Temperature and Heat Transfer

Q2

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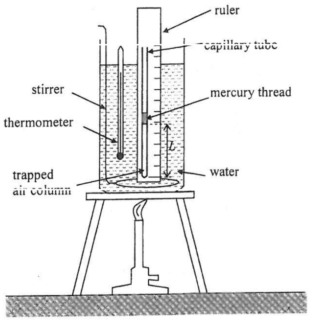

Figure 2.1

Figure 2.1 shows an air column trapped by a small mercury thread inside a uniform capillary tube. T set-up is heated by a water bath. The length of the air column L is measured at various temperature θ. Sor of the results are tabulated below:

Temperature θ / °C	20		92
Length of air column L / mm	64		80

(a) Describe the procedure(s) to be done before taking a reading in order to ensure that the trapped a reaches the same temperature as the water. (2 mark

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(b) Assume that length L increases linearly with temperature θ throughout.

(i) Estimate the length of the air column when the temperature indicated by the thermometer i 65 °C. (2 marks

(ii) Find the 'absolute zero' as obtained from this experiment. (2 marks)

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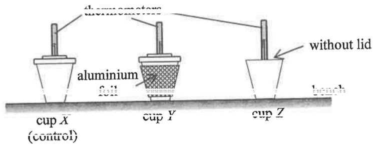
Q1

1. A student conducts an experiment to study how to best retain the warmth of water by using identical paper cups X, Y and Z shown in Figure 1.1. Each cup is filled with 250 cm³ of hot water and cup X serves as a control.

Figure 1.1

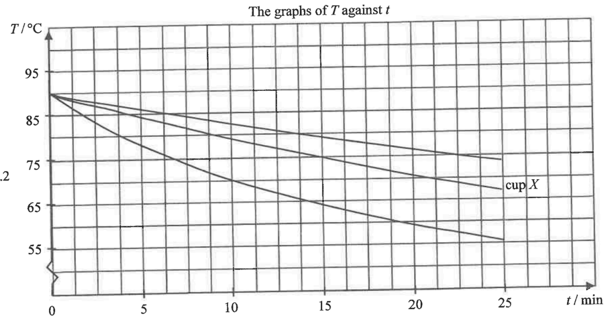


cup	wrapping	with a lid
X	no	yes
Y	aluminium wrapping	yes
Z	no	no

When the water temperature is 90 °C, the student starts taking thermometer readings every minute. Figure 1.2 shows the variation of temperature (T) of water in the cups with time (t) elapsed.

The graphs of T against t

Figure 1.2



(a) Suggest a reason why this experiment begins with the **same initial water temperature** (90 °C). (1 mark)

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(b) Explain why all the curves become less steep as time elapses. (2 marks)

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(c) (i) On Figure 1.2, indicate the respective curves that correspond to the results of **cup Y** and **cup Z**. (1 mark)

(ii) Explain your answer by referring respectively to the main heat transfer process. (3 marks)

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(d) Suggest a material for making cups so that heat loss via conduction can be reduced. (1 mark)

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